

Contents

1. Introduction to Graph Theory and Syllabus	2
1.1 What is a Graph? · 1.2 Remarks · 1.3 Applications · 1.4 Syllabus	2
2. Graph Terminology and Types of Graphs	4
2.1 Simple / Multigraph / Pseudograph · 2.2 Directed Graphs · 2.3 Adjacency & Incidence	4
3. Degree of a Vertex and the Handshaking Theorem	7
3.1 Degree · 3.2 Handshaking Theorem · 3.3 Even number of odd vertices	7
4. Special Types of Graphs	9
4.1 Complete · 4.2 Cycles · 4.3 Wheels · 4.4 Bipartite · 4.5 Complete Bipartite	9
5. Subgraphs and Operations on Graphs	13
6. Representing Graphs: Adjacency & Incidence Matrices	14
7. Isomorphism of Graphs	17
8. Connectivity: Paths, Circuits, Cut Vertices and Bridges	19
9. Euler Paths, Euler Circuits and Fleury's Algorithm	22
10. Hamiltonian Paths and Hamiltonian Circuits	24
11. Planar Graphs and Euler's Formula	26
12. Graph Colouring and the Chromatic Number	29
13. Question Bank — Unit III (Fully Solved)	31
14. Quick Revision Summary	40

1. Introduction to Graph Theory and Syllabus

1.1 What is a Graph?

A graph is the mathematical way of describing a set of objects together with the connections between them. Think of a metro map: the stations are points and the tracks between them are lines. Graph theory throws away all the geography (distances, directions, curves) and keeps only one thing — *which things are connected to which*.

Definition (Graph). A graph $G = (V, E)$ consists of a non-empty set V of **vertices** (or nodes) and a set E of **edges**. Each edge has one or two vertices associated with it, called its **endpoints**, and an edge is said to *connect* its endpoints.

The set V is called the vertex set and E the edge set. If an edge e joins vertices u and v , we write $e = \{u, v\}$ and say u and v are the endpoints of e .

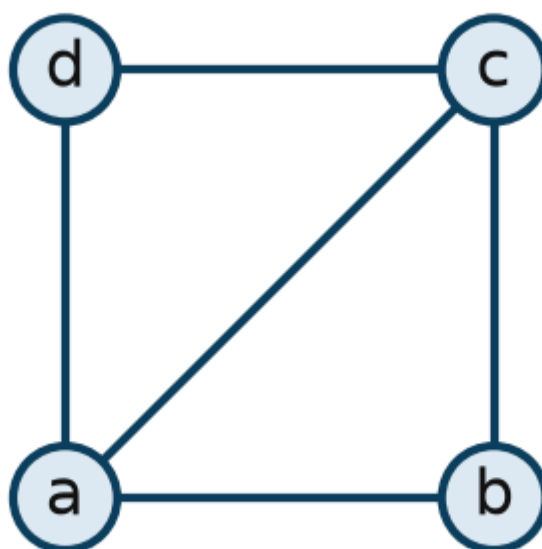


Fig. 1.1 — A graph with four vertices a, b, c, d and five edges: $\{a,b\}, \{b,c\}, \{c,d\}, \{d,a\}, \{a,c\}$.

1.2 Remarks on How We Draw a Graph

When we draw a picture of a graph we have a lot of freedom. All that matters is **the connections made by the edges**, not the particular geometry depicted. For example:

- the **lengths** of edges do not matter;
- whether edges **cross** each other does not matter (unless we specifically study *planar* graphs);
- how vertices are **placed or depicted** does not matter.

Two drawings that look completely different may represent the *same* graph. This is exactly the idea behind *isomorphism* (Section 7).

Finite vs Infinite. A graph with an infinite vertex set or infinitely many edges is an **infinite graph**. A graph with a finite vertex set and finitely many edges is a **finite graph**. Throughout this course we restrict our attention to finite graphs.

1.3 Why Graph Theory Matters (Applications)

Graphs are one of the most useful models in computer science and engineering, because a huge number of real problems are really "things connected to other things":

- **Computer networks** — routers (vertices) and links (edges); finding the shortest route for a packet is a graph problem.
- **Transportation & communication systems** — road maps, railway and airline networks.
- **Algorithms and data structures** — shortest-path (Dijkstra), minimum spanning tree (Kruskal/Prim), scheduling.
- **Chemistry** — molecules are graphs: atoms are vertices, chemical bonds are edges.
- **Circuit design** — components and wiring; **colouring** helps assign frequencies or gates.
- **Scheduling & resource allocation** — the "flight gates" problem (Section 12) is pure graph colouring.

1.4 Syllabus of Unit III (Graph Theory)

The topics covered in this unit, and where they appear in these notes:

- Introduction, terminology, types of graphs — **Sections 1-2**
- Degree, Handshaking theorem — **Section 3**
- Special graphs (K_n , C_n , W_n , bipartite, $K_{m,n}$) — **Section 4**
- Representations: adjacency & incidence matrices — **Section 6**
- Isomorphism — **Section 7**
- Paths, circuits, connectivity, cut vertices and bridges — **Section 8**
- Euler paths/circuits and Fleury's algorithm — **Section 9**
- Hamiltonian paths and circuits — **Section 10**
- Planar graphs and Euler's formula — **Section 11**
- Graph colouring, chromatic number, four-colour theorem — **Section 12**

2. Graph Terminology and Types of Graphs

2.1 Simple Graph, Multigraph and Pseudograph

Graphs are classified by what kind of edges they are allowed to have. The two "special" edge behaviours we care about are **loops** and **parallel edges**.

Loop. An edge that connects a vertex to itself is called a **loop**.

Simple Graph. In a simple graph each edge connects two **different** vertices, and no two edges connect the same pair of vertices. In other words, a simple graph has **no loops and no parallel (multiple) edges**.

Multigraph. A multigraph **may have multiple edges** connecting the same two vertices. When m different edges connect the vertices u and v , we say that $\{u, v\}$ is an edge of **multiplicity** m .

Pseudograph. A pseudograph may include **loops**, as well as **multiple edges** connecting the same pair of vertices.

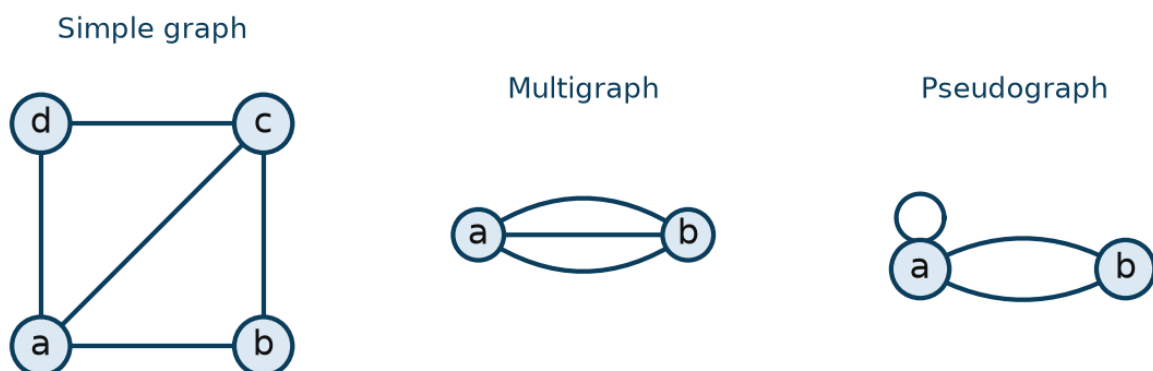


Fig. 2.1 — Left: a simple graph (no loops, no parallel edges). Centre: a multigraph (three parallel edges between a and b). Right: a pseudograph (a loop at a plus parallel edges).

How to remember it. Simple = "one clean edge per pair, no self-loops". Add parallel edges → **multigraph**. Also add loops → **pseudograph**. Every simple graph is a multigraph, and every multigraph is a pseudograph, but not the other way round.

2.2 Directed Graphs (Digraphs)

So far every edge was symmetric — if you can go from u to v you can also go from v to u . Many real situations are not like that (one-way streets, "follows" on social media, dependency between tasks). For these we use directed edges.

Directed Graph (Digraph). A directed graph $G = (V, E)$ consists of a non-empty set V of vertices and a set E of **directed edges** (or arcs). Each edge is associated with an **ordered pair** of vertices. The directed edge associated with the ordered pair (u, v) is said to **start at u** and **end at v** .

If (u, v) is an edge in G , then u is the **initial vertex** of the edge (and is adjacent *to* v), and v is the **terminal (end) vertex** (and is adjacent *from* u). The initial and terminal vertices of a loop are the same.

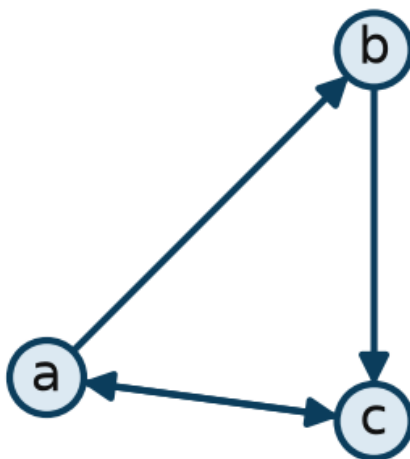


Fig. 2.2 — A directed graph with three vertices and four arcs: (a,b) , (b,c) , (c,a) , (a,c) .

In-degree and Out-degree. The **in-degree** of a vertex v , denoted $\deg^-(v)$, is the number of edges that **terminate at v** . The **out-degree** of v , denoted $\deg^+(v)$, is the number of edges with v as their **initial vertex**. A loop at a vertex contributes **1** to both the in-degree and the out-degree of that vertex.

Example. In Fig. 2.2: $\deg^-(a)=1$, $\deg^+(a)=2$; $\deg^-(b)=1$, $\deg^+(b)=1$; $\deg^-(c)=2$, $\deg^+(c)=1$. Check: total out-degree = $2+1+1 = 4$ = total in-degree = $1+1+2 = 4$ = number of arcs. ✓

Theorem (Digraph Handshaking). Let $G = (V, E)$ be a graph with directed edges. Then $\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|$.

Proof. The first sum counts the number of incoming edges over all vertices and the second sum counts the number of outgoing edges over all vertices. Every edge has exactly one tail and one head, so each edge is counted exactly once in each sum. Hence both sums equal the number of edges. $\square$

2.3 Adjacent Vertices, Incident Edges and Neighbourhood

Definition 1. Two vertices u, v in an undirected graph G are called **adjacent** (or **neighbours**) if there is an edge e between them. Such an edge e is called **incident** with the vertices u and v , and e is said to connect u and v .

Definition 2 (Degree). The **degree** of a vertex in an undirected graph is the number of edges incident with it, except that a **loop contributes two** to the degree of that vertex. The degree of the vertex v is denoted $\deg(v)$.

The **neighbourhood** $N(v)$ of a vertex v is the set of all vertices adjacent to it. An **isolated vertex** has degree 0, and a **pendant vertex** has degree 1.

3. Degree of a Vertex and the Handshaking Theorem

3.1 Working Out Degrees — A Full Example

The degree is simply "how many edge-ends touch this vertex". Count each incident edge once, and remember that a loop counts as **two**.

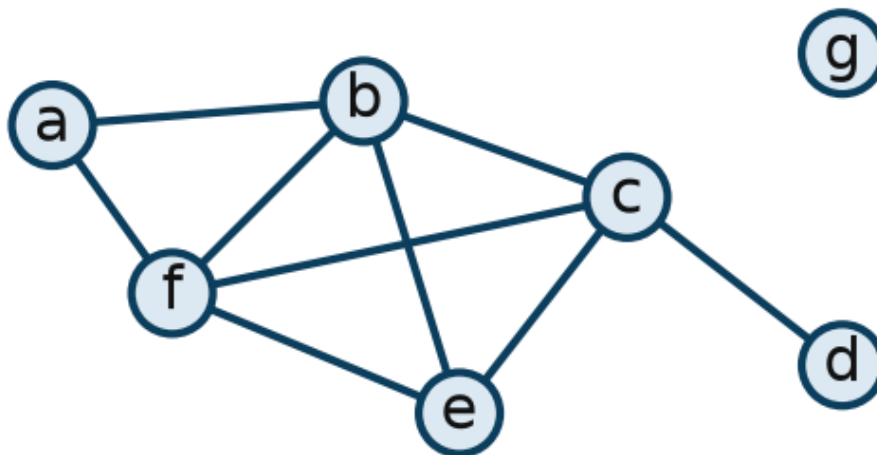


Fig. 3.1 — Graph G with vertices a, b, c, d, e, f, g , used for the degree example below.

Example. For the graph G in Fig. 3.1, find the degree and neighbourhood of every vertex.

Degrees:

$\deg(a)=2$, $\deg(b)=4$, $\deg(c)=4$, $\deg(d)=1$, $\deg(e)=3$, $\deg(f)=4$, $\deg(g)=0$.

Neighbourhoods:

$N(a)=\{b,f\}$, $N(b)=\{a,c,e,f\}$, $N(c)=\{b,d,e,f\}$, $N(d)=\{c\}$, $N(e)=\{b,c,f\}$, $N(f)=\{a,b,c,e\}$, $N(g)=\emptyset$.

Check with the Handshaking theorem: sum of degrees $=2+4+4+1+3+4+0=18=2 \times 9$, and the graph indeed has 9 edges. ✓

Notice that g is an **isolated vertex** ($\deg(g)=0$) and d is a **pendant vertex** ($\deg(d)=1$).

3.2 The Handshaking Theorem

Theorem 1 (Handshaking Theorem). If $G = (V, E)$ is an undirected graph with e edges, then $2e = \sum_{v \in V} \deg(v)$. That is, the sum of the degrees of all vertices equals **twice** the number of edges.

Proof. Each edge contributes exactly **twice** to the degree count of all vertices — once at each of its two endpoints (a loop contributes 1 at each "end", i.e. 2 total). Hence both the left-hand side ($2e$) and the right-hand side (the sum of degrees) count the total number of edge-ends, so they are equal. $\square$

Example 1. How many edges are there in a graph with 10 vertices, each of degree 6?

Sum of degrees = $6 \times 10 = 60$. By the Handshaking theorem, $2e = 60$, so $e = 30$.

Example 2. If a graph has 5 vertices, can each vertex have degree 3?

If so, the sum of degrees would be $3 \times 5 = 15$, which is **odd**. But by the theorem the sum must equal $2e$, which is always **even**. Hence this is impossible — such a graph cannot exist.

3.3 The Number of Odd-Degree Vertices is Even

Theorem 2. The number of vertices of **odd degree** in any undirected graph is always **even**.

Proof. Let V_1 be the set of vertices of even degree and V_2 the set of vertices of odd degree in an undirected graph $G = (V, E)$ with m edges. We know $\sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v) = 2m$. The left side $2m$ is even. The first sum $\sum_{v \in V_1} \deg(v)$ is also even (sum of even numbers). Therefore the second sum $\sum_{v \in V_2} \deg(v)$ must be even. But this sum is a sum of odd numbers, and a sum of odd numbers is even only when there is an **even number of terms**. Hence the number of vertices of odd degree is even. $\square$

Why this theorem is powerful. It instantly rules out many "impossible" graphs. E.g. a graph cannot have exactly 3 vertices of odd degree, or exactly 1 vertex of odd degree. This fact is also the key to the existence of Euler paths (Section 9).

4. Special Types of Graphs

Certain graphs appear so often that they have their own names and notation. You should be able to draw them and quote their vertex/edge counts instantly.

4.1 Complete Graphs K_n

Complete Graph. The complete graph on n vertices, denoted K_n , is the simple graph that contains **exactly one edge between each pair of distinct vertices**.

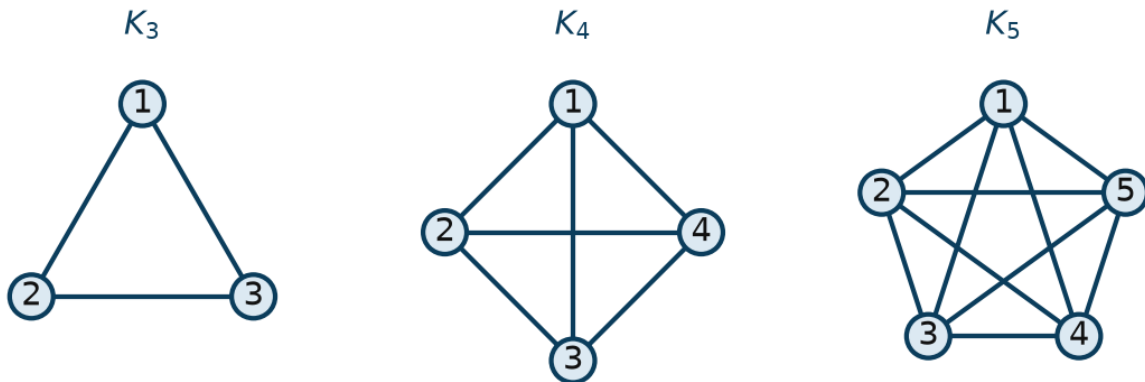


Fig. 4.1 — Complete graphs K_3 , K_4 , K_5 .

Since every pair of vertices is joined, the number of edges is the number of ways of choosing 2 vertices out of n :

Key formula. $|E(K_n)| = C(n, 2) = n(n-1)/2$, and every vertex has degree $n-1$. So K_5 has 10 edges, K_{100} has 4950 edges.

4.2 Cycle Graphs C_n

Cycle. A cycle C_n for $n \geq 3$ consists of n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}, \{v_n, v_1\}$.

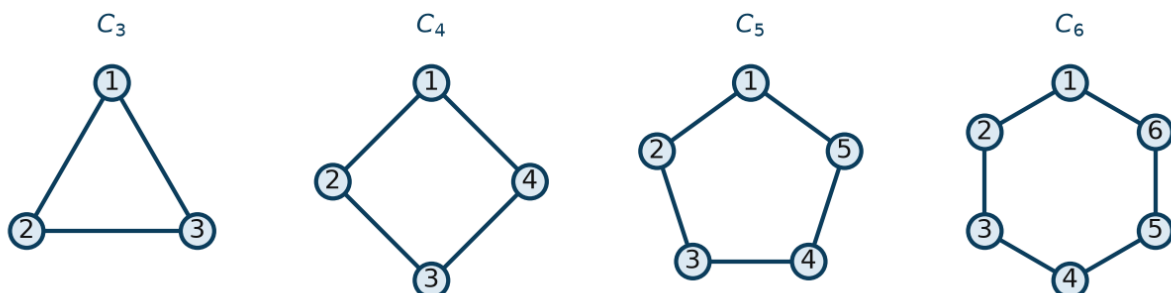


Fig. 4.2 — Cycles C_3 , C_4 , C_5 , C_6 . Each has n vertices and n edges, and every vertex has degree 2.

4.3 Wheel Graphs W_n

Wheel. A wheel W_n is obtained by taking a cycle C_n ($n \geq 3$) and adding one extra vertex (the **hub**), joining it to each of the n vertices of the cycle by new edges.

Wheel W_5

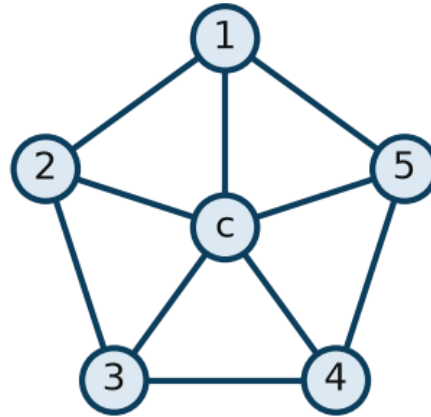


Fig. 4.3 — The wheel W_5 : a C_5 plus a central hub joined to all 5 rim vertices.

W_n has $n+1$ vertices and $2n$ edges; the hub has degree n and each rim vertex has degree 3.

4.4 Bipartite Graphs

Bipartite Graph. A simple graph G is **bipartite** if its vertex set V can be partitioned into two disjoint subsets V_1 and V_2 such that **every edge connects a vertex in V_1 to a vertex in V_2** . In other words, there are no edges that join two vertices within V_1 or within V_2 .

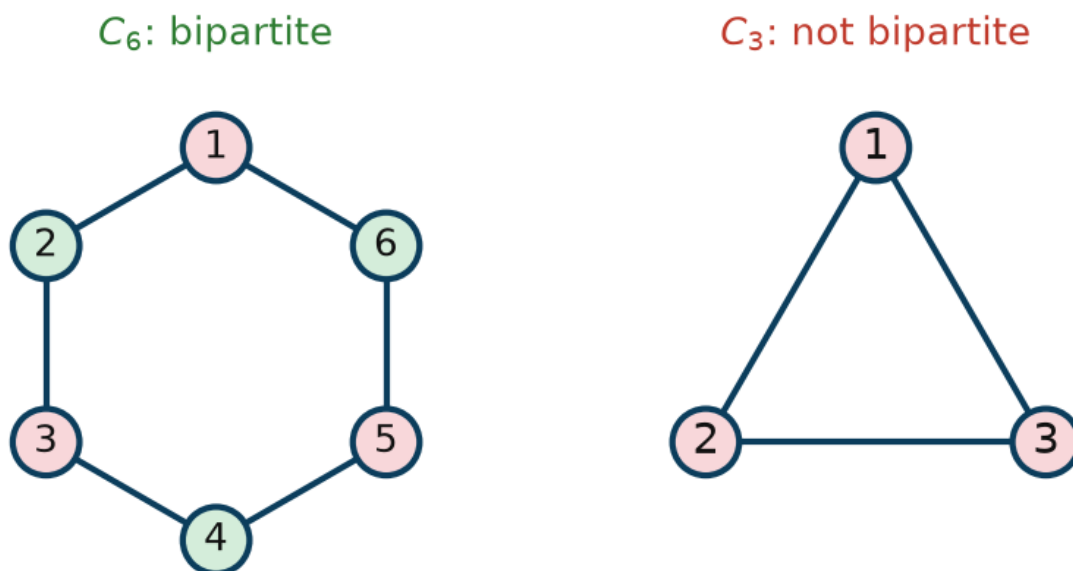


Fig. 4.4 — C_6 is bipartite (split $\{v_1, v_3, v_5\}$ and $\{v_2, v_4, v_6\}$); C_3 (a triangle) is **not** bipartite.

Example. Show that C_6 is bipartite and C_3 is not.

C_6 : partition the vertex set as $V_1 = \{v_1, v_3, v_5\}$ and $V_2 = \{v_2, v_4, v_6\}$. Every edge of C_6 joins a vertex of V_1 to one of V_2 , so C_6 is bipartite.

C_3 : a triangle has three vertices all mutually adjacent, so any 2-way split would put two adjacent vertices on the same side. Hence C_3 is not bipartite.

Theorem. A graph is **bipartite if and only if it contains no odd cycle** (no cycle of odd length).

Practical test. Try to 2-colour the graph (say red/green) so that neighbours differ. If you succeed, it is bipartite; if some vertex is forced to be both colours, there is an odd cycle and it is not bipartite.

4.5 Complete Bipartite Graphs $K_{m,n}$

Complete Bipartite Graph. A complete bipartite graph $K_{m,n}$ has its vertex set partitioned into two subsets V_1 of size m and V_2 of size n , such that there is an edge from **every** vertex in V_1 to **every** vertex in V_2 .

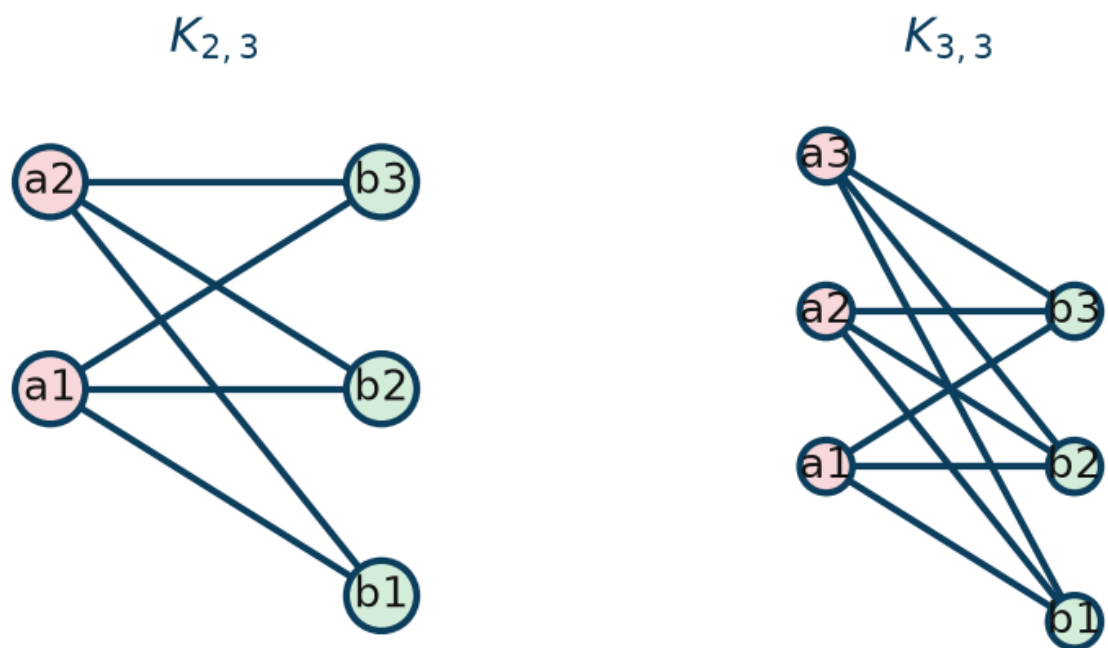


Fig. 4.5 — Complete bipartite graphs $K_{2,3}$ and $K_{3,3}$.

Key formula. $|E(K_{m,n})| = m \times n$. So $K_{2,6}$ has $2 \times 6 = 12$ edges and $K_{3,3}$ has 9 edges. $K_{3,3}$ is the famous graph used to prove non-planarity (Section 11).

5. Subgraphs and Operations on Graphs

Subgraph. A **subgraph** of a graph $G = (V, E)$ is a graph $H = (W, F)$ where $W \subseteq V$ and $F \subseteq E$. A subgraph H is a **proper subgraph** of G if $H \neq G$.

In words: pick some vertices, and pick some of the edges that join them — that is a subgraph. A subgraph obtained by deleting only vertices (and all edges touching them) is called an **induced subgraph**.

Union of Graphs. The union of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the simple graph with vertex set $V_1 \cup V_2$ and edge set $E_1 \cup E_2$. It is denoted $G_1 \cup G_2$.

Example. If G_1 and G_2 are two triangles sharing a single vertex, then in each of G_1, G_2 every edge is a cut edge, but in the union $G_1 \cup G_2$ no edge is a cut edge — because there is now an alternative route around. The shared vertex is a cut vertex of all three graphs.

6. Representing Graphs: Adjacency & Incidence Matrices

To feed a graph to a computer we must store it as numbers. The two standard encodings are the adjacency matrix and the incidence matrix.

6.1 Adjacency List

Adjacency List. An adjacency list represents a graph (with no multiple edges) by listing, for each vertex, the vertices adjacent to it.

For the graph in Fig. 6.1: a: b, c; b: a, d; c: a, d; d: b, c. This uses space proportional to the number of edges — good for sparse graphs.

6.2 Adjacency Matrix

Adjacency Matrix. Let $G = (V, E)$ be a graph with vertices ordered as $v_1, v_2, \dots, v_n$. The adjacency matrix is the $n \times n$ matrix $A_G = [a_{ij}]$ where $a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G, \\ 0 & \text{otherwise.} \end{cases}$

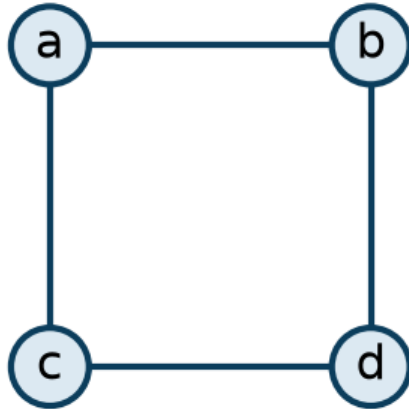


Fig. 6.1 — Graph on vertices a, b, c, d with edges $\{a, b\}, \{b, d\}, \{d, c\}, \{c, a\}$ (a 4-cycle $a \setminus b \setminus d \setminus c \setminus a$).

Example. The adjacency matrix of Fig. 6.1 (ordering a, b, c, d) is $A = \begin{vmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{vmatrix}$. Note

it is **symmetric** for an undirected graph.

Loops and multiple edges. The adjacency matrix can also represent pseudographs: a **loop** at v_i gives a 1 in the (i,i) position, and **parallel edges** make the (i,j) entry equal to the number of edges joining v_i and v_j .

6.3 Incidence Matrix

Incidence Matrix. Let $G = (V,E)$ be an undirected graph with vertices $v_1, \dots, v_n$ and edges $e_1, \dots, e_m$. The incidence matrix is the $n \times m$ matrix $M = [m_{ij}]$ where $m_{ij} =$

$\begin{cases} 1 & \text{if edge } e_j \text{ is incident with } v_i, \\ 0 & \text{otherwise.} \end{cases}$ For a loop, the entry is 2 (both ends touch the same vertex).

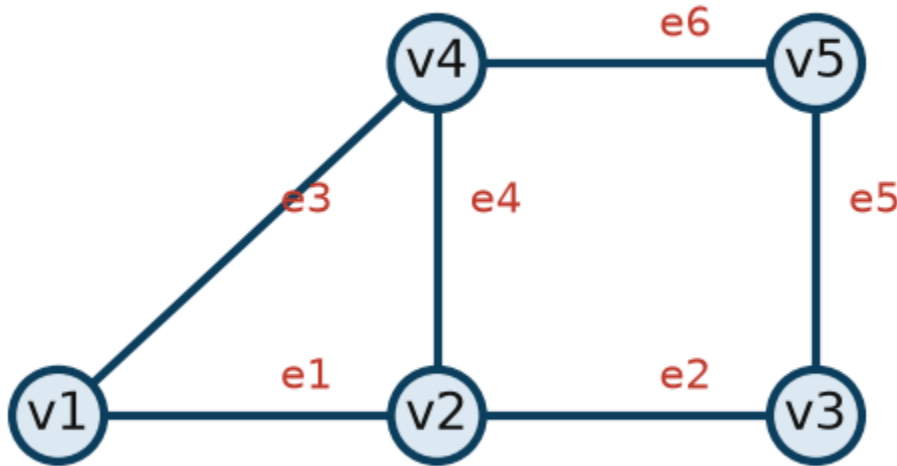


Fig. 6.2 — Graph with vertices $v_1, \dots, v_5$ and edges $e_1, \dots, e_6$.

Example. The incidence matrix of Fig. 6.2 is (rows $v_1..v_5$, columns $e_1..e_6$): $M =$

$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$. Each **column** has exactly two 1s (each edge touches two vertices),

and the sum of each **row** is the degree of that vertex.

6.4 Counting Paths Using the Adjacency Matrix

Theorem. Let G be a graph with adjacency matrix A (vertices ordered $v_1, \dots, v_n$). Then the (i,j) entry of A^k equals the **number of paths of length exactly k from v_i to v_j** .

Example. For Fig. 6.1, the number of paths of length 4 from a to d is the (1,4) entry of A^4 . Computing gives 8, with the eight paths being a,b,a,b,d; a,b,a,c,d; a,b,d,b,d; a,b,d,c,d; a,c,a,b,d; a,c,a,c,d; a,c,d,b,d; a,c,d,c,d. This "matrix power" trick is the standard tool for path-counting questions.

7. Isomorphism of Graphs

Two graphs may look completely different when drawn, yet be "the same graph" underneath — just with the vertices renamed and rearranged. That idea is **isomorphism**.

Isomorphic Graphs. The simple graphs $G_1=(V_1,E_1)$ and $G_2=(V_2,E_2)$ are **isomorphic** if there is a bijective (one-to-one and onto) function $f: V_1 \rightarrow V_2$ such that a and b are adjacent in G_1 **if and only if** $f(a)$ and $f(b)$ are adjacent in G_2 , for all $a,b \in V_1$. Such f is called an **isomorphism**. Two graphs that are not isomorphic are **non-isomorphic**.

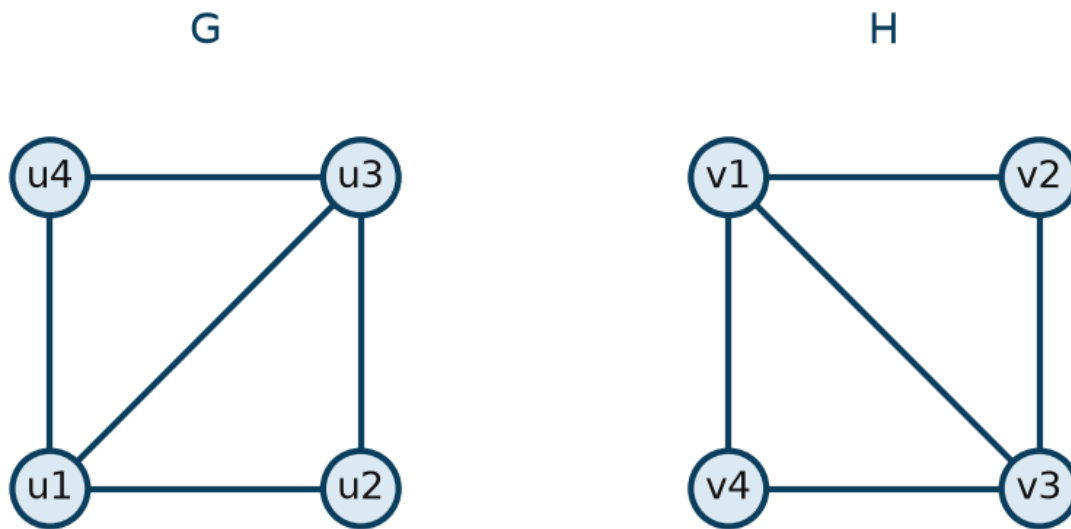


Fig. 7.1 — G and H are isomorphic: the map $f(u_1)=v_1, f(u_2)=v_4, f(u_3)=v_3, f(u_4)=v_2$ preserves all adjacencies.

Example. Show that the graphs G and H in Fig. 7.1 are isomorphic.

The function f with $f(u_1)=v_1, f(u_2)=v_4, f(u_3)=v_3, f(u_4)=v_2$ is a one-to-one correspondence. The adjacent pairs in G are $\{u_1,u_2\}, \{u_1,u_3\}, \{u_1,u_4\}, \{u_2,u_3\}$. Their images under f are $\{v_1,v_4\}, \{v_1,v_3\}, \{v_4,v_2\}, \{v_3,v_2\}$, and each of these is an edge of H . So f is an isomorphism and $G \cong H$.

7.1 How to Show Two Graphs Are *Not* Isomorphic (Graph Invariants)

Checking all $n!$ possible bijections is impractical. Instead we use **graph invariants** — properties preserved by isomorphism. If two graphs differ on any invariant, they cannot be isomorphic.

Useful invariants.

- the **number of vertices**;
- the **number of edges**;
- the **degree sequence** (the multiset of degrees, written in non-increasing order);
- the number of vertices of each degree; number of cycles of each length; whether the graph is bipartite/connected, etc.

Example. Two graphs each have 8 vertices and 10 edges, and both have four vertices of degree 2 and four of degree 3. Are they isomorphic?

Not necessarily. In the first graph, a degree-2 vertex a is adjacent to vertices of degrees 1 and 3; but in the second graph every degree-2 vertex is adjacent to another degree-2 vertex. This difference in *how degrees are arranged* (not just the degree sequence) proves the graphs are **not** isomorphic.

Complexity note. The best known algorithms for graph isomorphism have exponential worst-case complexity in the number of vertices, so brute force is infeasible for large n .

8. Connectivity: Paths, Circuits, Cut Vertices and Bridges

8.1 Paths and Circuits

Path. A **path** is a sequence of edges from one vertex to another (equivalently, a sequence of vertices in a simple graph). A path is a **circuit** if it begins and ends at the same vertex. A path or circuit is **simple** if it does not contain the same edge more than once.

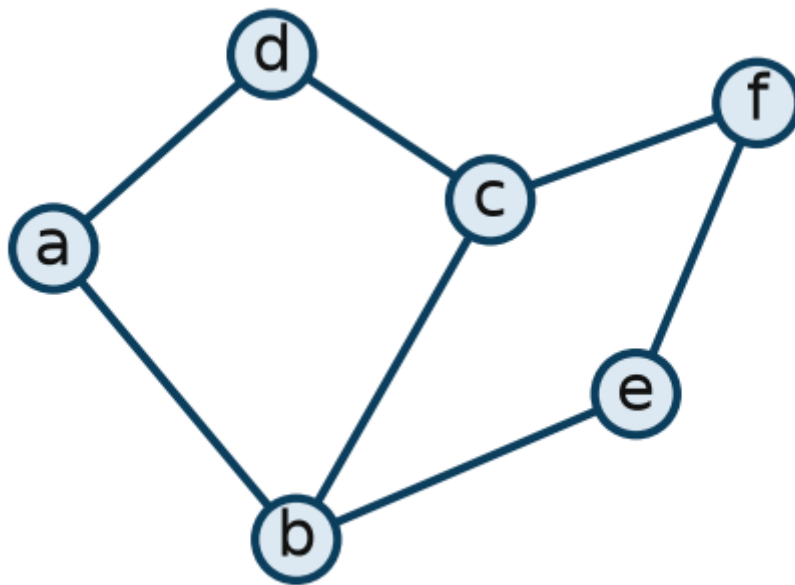


Fig. 8.1 — A graph used to illustrate paths and circuits.

Example (using Fig. 8.1).

- a, d, c, f, e is a simple path of length 4.
- b, c, f, e, b is a circuit of length 4.
- a, b, e, d, a, b is a path of length 5, but it is **not** simple (the edge {a,b} is repeated).

8.2 Connected and Disconnected Graphs

Connected. An undirected graph is **connected** if there is a path between every pair of vertices. A graph that is not connected is **disconnected**. We **disconnect** a graph when we remove vertices or edges (or both) to produce a disconnected subgraph.

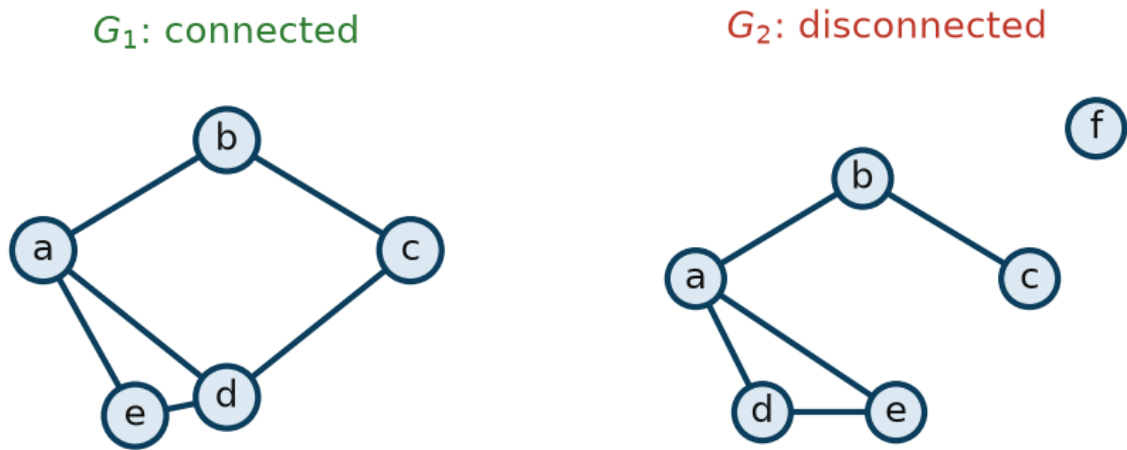


Fig. 8.2 — G_1 is connected (a path exists between every pair). G_2 is disconnected (e.g. there is no path between a and f).

8.3 Cut Vertices and Bridges

Cut Vertex (Articulation Point). If removing a vertex (and all its incident edges) produces a graph with more components, that vertex is a **cut vertex**.

Cut Edge (Bridge). If removing an edge creates more components, that edge is a **cut edge** or **bridge**.

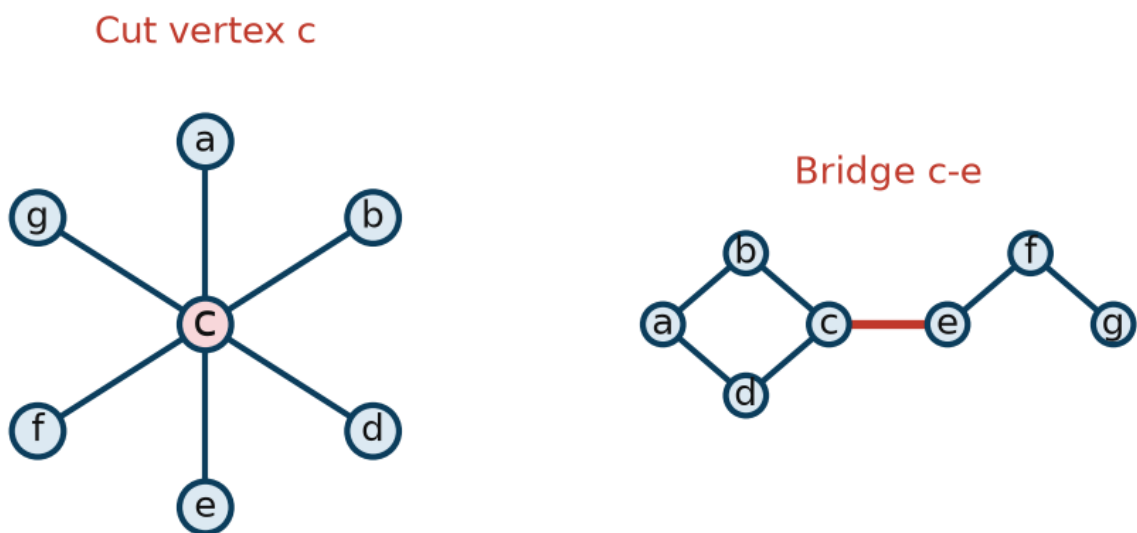


Fig. 8.3 — Left: in the star network the centre c is a cut vertex (all edges are cut edges). Right: the edge c - e is a bridge joining two components.

Example. In Fig. 8.2's G_1 every edge is a cut edge, but if we take the union of two such graphs sharing vertex e , then no edge is a cut edge (there is an alternative route). The shared vertex e is a cut vertex in all of them.

8.4 Counting Paths Between Vertices

As seen in Section 6.4, the number of paths of length k from v_i to v_j is the (i,j) entry of A^k . For Fig. 6.1 there are 8 paths of length 4 from a to d .

9. Euler Paths, Euler Circuits and Fleury's Algorithm

The famous origin of graph theory: the **Königsberg bridge problem** — can you walk through the city crossing every bridge exactly once? Euler answered "no", and in doing so invented graph theory.

Euler Circuit and Euler Path. An **Euler circuit** in a graph G is a simple circuit that contains **every edge** of G . An **Euler path** in G is a simple path that contains **every edge** of G .

So an Euler circuit starts and ends at the same vertex and traverses every edge exactly once; an Euler path traverses every edge exactly once but need not return to its start.

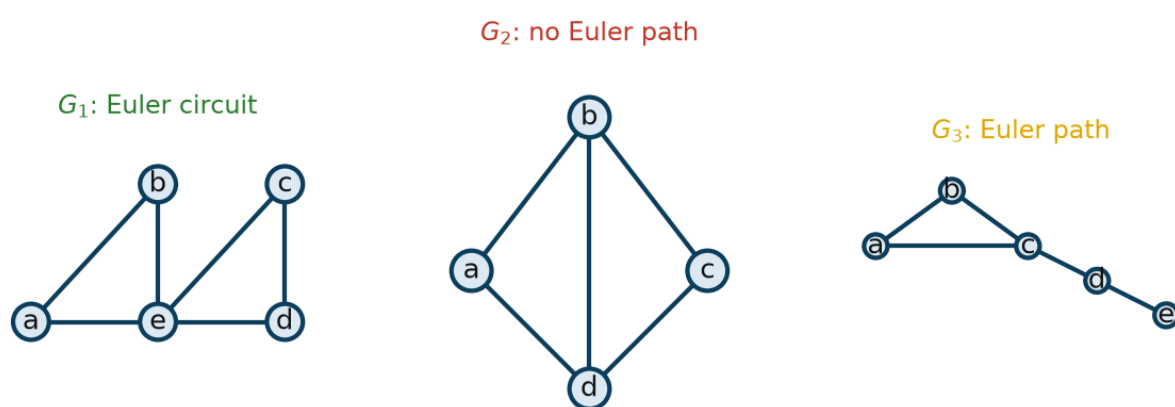


Fig. 9.1 — G_1 has an Euler circuit; G_2 has neither; G_3 has an Euler path but no Euler circuit.

Example. Which of G_1 , G_2 , G_3 in Fig. 9.1 has an Euler circuit? Of those that do not, which has an Euler path?

G_1 has an Euler circuit, e.g. a, e, c, d, e, b, a . Neither G_2 nor G_3 has an Euler circuit. G_3 has an Euler path, e.g. a, c, d, e, b, d, a, b , but G_2 has no Euler path.

Necessary and Sufficient Conditions.

- A connected multigraph has an **Euler circuit** if and only if **every vertex has even degree**.
- A connected multigraph has an **Euler path but not an Euler circuit** if and only if it has **exactly two vertices of odd degree** (the path must start at one odd vertex and end at the other).
- If a connected graph has **more than two** vertices of odd degree, it has **neither**.

Why? Every time you pass *through* a vertex in a circuit you use one edge to enter and one to leave — so edges come in pairs, forcing even degree. Only the start/end vertex of an Euler path can have an odd count.

9.1 Fleury's Algorithm

Once we know an Euler path/circuit exists, Fleury's algorithm tells us *how to actually trace it*. The key idea is: **do not burn your bridges** — never cross a bridge (cut edge) unless there is no other choice.

Fleury's Algorithm (Euler circuit).

1. **Check the conditions:** the graph must be connected and every vertex must have even degree.
2. **Start at any vertex** (allowed because all degrees are even).
3. **Traverse:** from the current vertex, move along any edge that is **not a bridge**, unless no other edge remains.
4. **Remove** each traversed edge so it is never reused.
5. **Repeat** until all edges are traversed; you will return to the starting vertex.

For an **Euler path**, start instead at one of the **two odd-degree** vertices; the rest is the same.

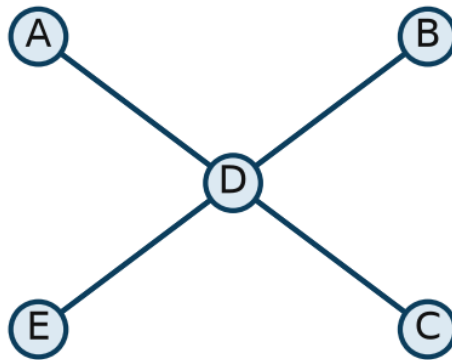


Fig. 9.2 — Graph for Fleury's algorithm; all vertices have even degree, so an Euler circuit exists.

Worked example (Fig. 9.2). Find an Euler circuit starting at A.

1. Start at A; choose edge A-D.
2. From D the choices are D-B, D-C, D-E. Choosing D-C would disconnect the graph (it is a bridge), so we choose D-E.
3. From E there is only one option (E-B); the rest of the circuit is then forced.

Circuit obtained: $\{A \rightarrow D \rightarrow E \rightarrow B \rightarrow D \rightarrow C \rightarrow A\}$.

10. Hamiltonian Paths and Hamiltonian Circuits

Euler asked for a walk that covers every **edge** once. Hamilton asked a different question: is there a walk that visits every **vertex** once?

Hamiltonian Circuit. A **Hamiltonian circuit** (or Hamiltonian cycle) is a cycle in a connected graph that contains **all the vertices** of the graph. A graph that contains a Hamiltonian circuit is called a **Hamiltonian graph**.

Hamiltonian Path. A **Hamiltonian path** is a path in a connected graph that contains **all the vertices** of the graph (edges may be missed, but no edge is repeated). A Hamiltonian circuit is exactly a Hamiltonian path that starts and ends at the same vertex.

Hamiltonian circuit (green)

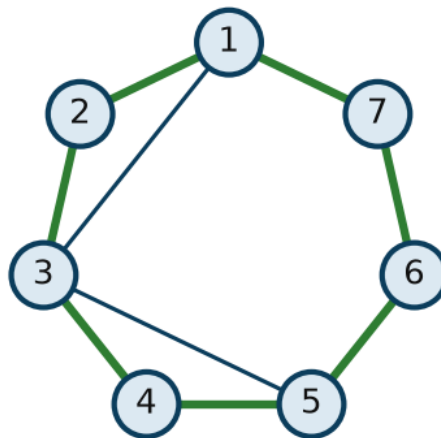


Fig. 10.1 — The green closed walk visits every vertex exactly once (except the start): a Hamiltonian circuit.

Example. The graph in Fig. 10.1 contains the closed walk $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow A$, which visits every vertex exactly once (except the starting vertex) and repeats no edge. Hence it is a Hamiltonian graph.

Important notes.

- Any Hamiltonian circuit becomes a Hamiltonian path by deleting one of its edges.
- Every graph that has a Hamiltonian circuit also has a Hamiltonian path, but **not vice versa**.
- A graph may contain **more than one** Hamiltonian path/circuit.
- Unlike Euler's case, there is **no simple necessary-and-sufficient condition** for a Hamiltonian circuit — deciding it is a hard (NP-complete) problem. We only have sufficient conditions, e.g. **Dirac's theorem**: if every vertex of a graph with $n \geq 3$ vertices has degree $\geq n/2$, then the graph is Hamiltonian.

11. Planar Graphs and Euler's Formula

Planar Graph. A **planar graph** is a graph that can be drawn in a plane so that **none of its edges cross** each other. Such a drawing is called a planar representation.

Note: a graph is planar if *some* crossing-free drawing exists — even if the drawing we first see has crossings.

11.1 Regions of a Plane Graph

Regions. A planar representation of a graph splits the plane into connected areas called **regions** (or faces). Each region has a degree:

- degree of an **interior region** = number of edges enclosing it;
- degree of the **exterior (outer) region** = number of edges exposed to it.

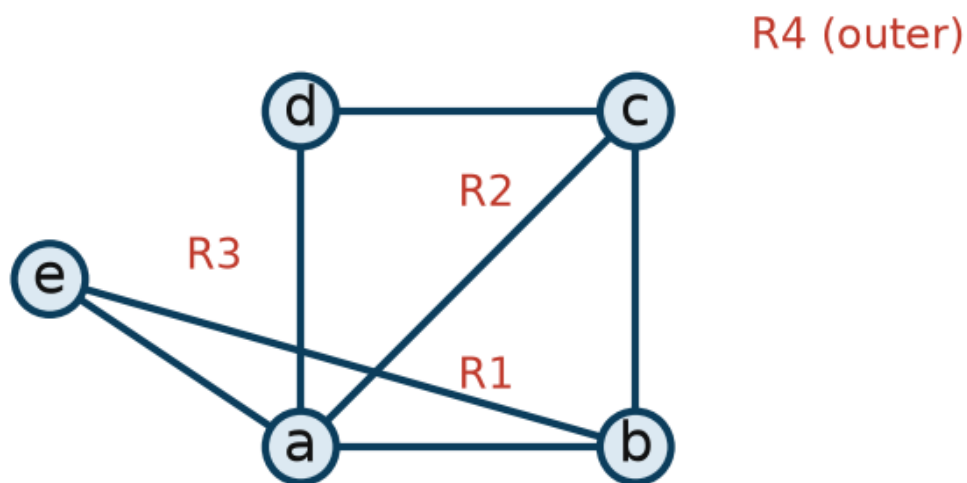


Fig. 11.1 — A planar graph splitting the plane into four regions R_1 , R_2 , R_3 (interior) and R_4 (exterior), with region degrees 3, 3, 3, 5.

Example (Fig. 11.1). The planar graph splits the plane into 4 regions where $\deg(R_1)=3$, $\deg(R_2)=3$, $\deg(R_3)=3$, $\deg(R_4)=5$. Note the sum of region-degrees $= 3+3+3+5=14=2 \times 7=2e$. ✓

Two counting properties.

- **Property 1:** sum of degrees of all **vertices** $= 2 \times (\text{number of edges})$.
- **Property 2:** sum of degrees of all **regions** $= 2 \times (\text{number of edges})$ — because each edge borders exactly two regions (possibly the same region twice).

11.2 Euler's Formula for Planar Graphs

Euler's Formula. For a **connected** planar graph with v vertices, e edges and r regions, $v - e + r = 2$. More generally, if the planar graph has k components, then $v - e + r = k + 1$.

Problem 1. Let G be a connected planar simple graph with 25 vertices and 60 edges. Find the number of regions.

Solution. Given $v=25$, $e=60$. By Euler's formula, $r = e - v + 2 = 60 - 25 + 2 = 37$. So G has **37 regions**.

Problem 2. Let G be a planar graph with 10 vertices, 3 components and 9 edges. Find the number of regions.

Solution. Given $v=10$, $e=9$, $k=3$. Using $r = e - v + (k+1) = 9 - 10 + 4 = 3$. So G has **3 regions**.

11.3 Proof of Euler's Formula (by Induction)

Proof. Base case: for a single edge, $e_1=1$, $v_1=2$, $r_1=1$, and $v-e+r = 2-1+1 = 2$. ✓

Inductive step: assume $r_n = e_n - v_n + 2$ holds for a graph G_n , and add one edge $\{a_{n+1}, b_{n+1}\}$ to get G_{n+1} . There are two cases:

Case 1 (the new edge joins two existing vertices and splits a region): then $r_{n+1}=r_n+1$, $e_{n+1}=e_n+1$, $v_{n+1}=v_n$, so $r_{n+1} = e_{n+1} - v_{n+1} + 2$ still holds.

Case 2 (the new edge adds a new vertex, i.e. a pendant edge): then $r_{n+1}=r_n$, $e_{n+1}=e_n+1$, $v_{n+1}=v_n+1$, and again $r_{n+1} = e_{n+1} - v_{n+1} + 2$. Hence the formula holds for all connected planar graphs. □

11.4 Corollaries and Bounds

Corollary 1. Let G be a connected simple planar graph with v vertices, $e > 2$ edges and r regions. Then $3r \leq 2e$ and $e \leq 3v - 6$.

Proof. Since G is simple (loop-free, no multiple edges), the boundary of every region (including the outer one) contains at least three edges, so every region has degree ≥ 3 . Each edge occurs exactly twice on region boundaries, so $2e \geq 3r$, i.e. $r \leq 2/3e$. Substituting into $v-e+r=2$: $2 \leq v - e + 2/3e = v - e/3$, giving $6 \leq 3v - e$, i.e. $e \leq 3v - 6$. □

Corollary 2. Every connected simple planar graph has a vertex of degree at most 5.

Proof. If G has one or two vertices the result is clear. Otherwise, by Corollary 1, $e \leq 3v - 6$, so $2e \leq 6v - 12$. Suppose, for contradiction, that every vertex had degree ≥ 6 . Then by the Handshaking theorem, $2e = \sum \deg(v) \geq 6v$. But this contradicts $2e \leq 6v - 12$. Hence at least one vertex has degree ≤ 5 . $\square$

Corollary 3. If G is a connected simple planar graph with $v \geq 3$ vertices, e edges and **no triangles** (no circuits of length 3), then $e \leq 2v - 4$.

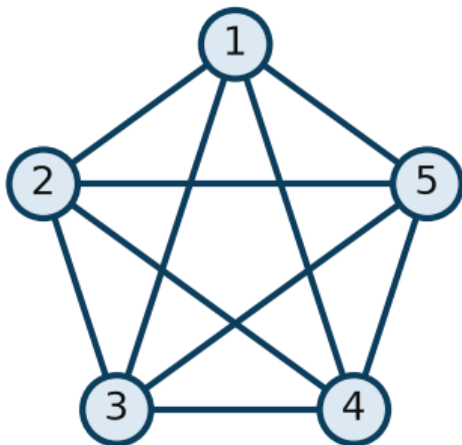
Idea: same as Corollary 1, except that "no triangles" forces every region to have degree ≥ 4 , giving $2e \geq 4r$, hence $e \leq 2v - 4$.

Example — K_5 and $K_{3,3}$ are not planar.

For K_5 : $v=5$, $e=10$, but $3v-6 = 9 < 10$. Since K_5 violates $e \leq 3v-6$, it cannot be planar.

For $K_{3,3}$: it has $v=6$, $e=9$ and no triangles, so $e \leq 2v-4 = 8$ would be required, but $9 > 8$. Hence $K_{3,3}$ is not planar either.

K_5 : not planar



$K_{3,3}$: not planar

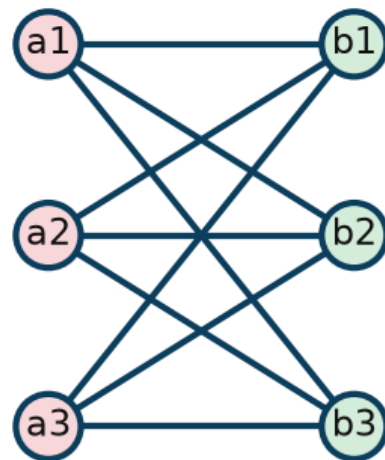


Fig. 11.2 — K_5 and $K_{3,3}$: the two basic non-planar graphs (Kuratowski's theorem says every non-planar graph contains a subdivision of one of these).

12. Graph Colouring and the Chromatic Number

Graph Colouring Problem. Given a graph, colour all the vertices so that **two adjacent vertices get different colours**. The objective is to use the **minimum** number of colours.

Chromatic Number. The minimum number of colours needed to properly colour a graph G is called its **chromatic number**, denoted $\chi(G)$.

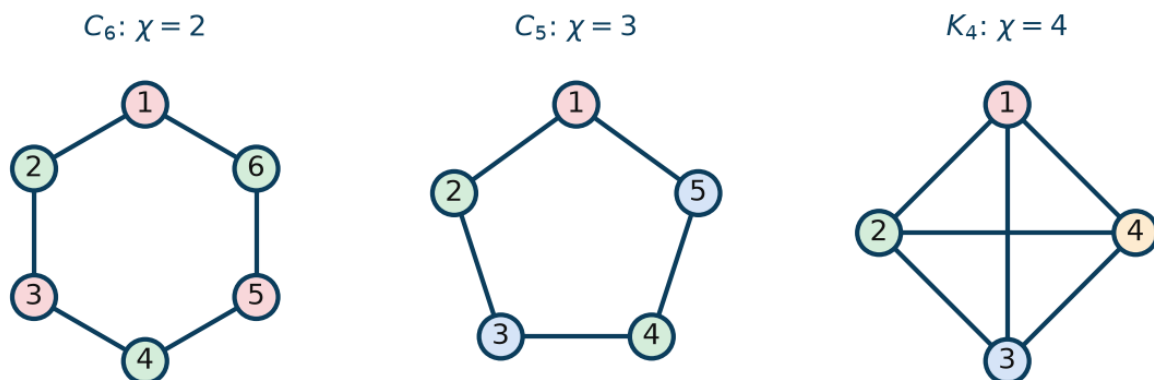


Fig. 12.1 — C_6 needs 2 colours, C_5 needs 3, and K_4 needs 4.

12.1 Chromatic Numbers of Common Graphs

Learn these by heart.

- $\chi(G) = 1$ iff G has **no edges**.
- $\chi(\text{path}) = 2$, and $\chi(\text{tree}) = 2$ (bipartite).
- $\chi(C_n) = 2$ if n is **even**, and $\chi(C_n) = 3$ if n is **odd** (an odd cycle needs 3 colours).
- $\chi(K_n) = n$ (every pair is adjacent, so all n colours are needed).
- $\chi(W_n) = 4$ when n is odd, and $\chi(W_n) = 3$ when n is even.

Example. $\chi(K_5) = 5$ (all five vertices mutually adjacent). $\chi(C_7) = 3$ (odd cycle). $\chi(K_4) = 4$.

12.2 Bipartite and 2-Colourability

Theorem. A graph is **2-colourable if and only if it is bipartite**, i.e. if and only if it has **no odd cycle**.

How to 2-colour: pick any vertex v and colour it red; colour all its neighbours green; colour all neighbours of green vertices red; and so on. If this process ever forces a vertex to receive two colours, the graph contains an odd cycle and is not bipartite.

12.3 Colouring Planar Graphs — The Four-Colour Theorem

The Four-Colour Theorem. For every **planar** graph, the chromatic number is at most 4: $\chi(G) \leq 4$.

This was posed as a conjecture in the 1850s and finally proved in 1976 by Appel and Haken with the aid of computers. It says that any map can be coloured with at most four colours so that neighbouring regions differ — the map's adjacency graph is planar.

12.4 Application: Assigning Flight Gates

Flights need gates, but some flights overlap in time and cannot share a gate. Build a **conflict graph**: each vertex is a flight, and an edge joins two flights that need a gate at the same time. Then a valid gate assignment using k gates exists **iff** the conflict graph has a k -colouring. Minimising the number of gates is exactly finding the chromatic number — a direct real-world use of graph colouring.

13. Question Bank — Unit III (Fully Solved)

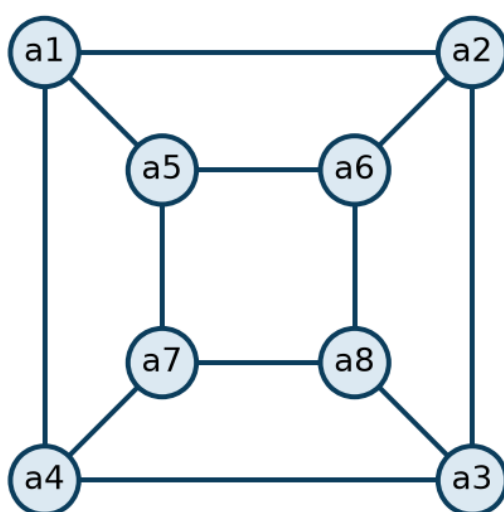
Each solution states what is asked, the idea/tool used and why, the steps, and the final boxed answer.

Q1. Define a bipartite graph. Check whether the graph shown is bipartite.

Definition: A simple graph is bipartite if its vertex set can be split into two disjoint sets V_1, V_2 such that every edge joins a vertex of V_1 to a vertex of V_2 (no edge lies inside one part).

Test and why: A graph is bipartite iff it has no odd cycle. Equivalently, try a 2-colouring — colour one vertex red, all its neighbours green, all their neighbours red, and so on; if the colouring completes with no conflict, the graph is bipartite.

G (8 vertices, all degree 3)



Graph for Q1 — 8 vertices, every vertex of degree 3 (a cube graph).

Apply: Take $V_1 = \{a_1, a_3, a_6, a_7\}$ and $V_2 = \{a_2, a_4, a_5, a_8\}$. Every edge — the outer cycle $a_1 a_2 a_3 a_4$, the inner cycle $a_5 a_6 a_8 a_7$ and the spokes $a_1 a_5, a_2 a_6, a_3 a_8, a_4 a_7$ — joins a vertex of V_1 to one of V_2 .

The graph is bipartite

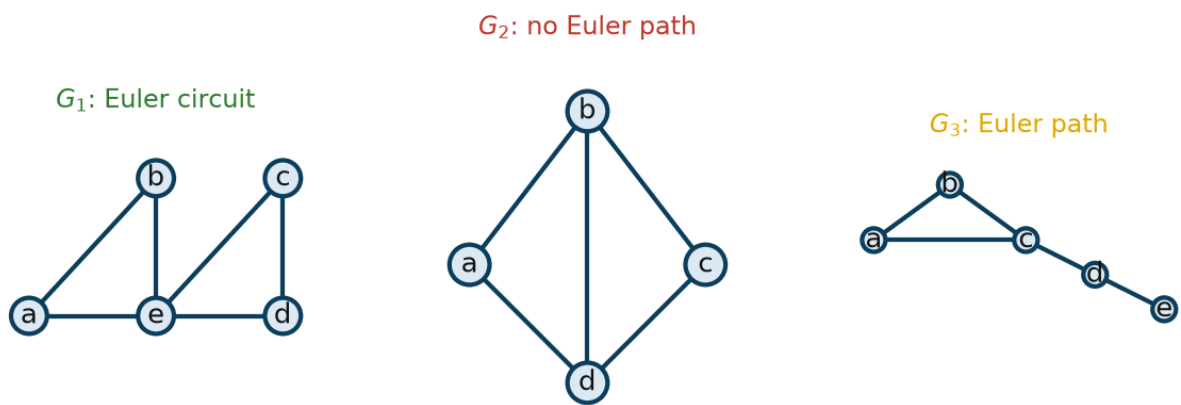
Q2. Define an Euler circuit. State the necessary and sufficient condition for a connected graph to have an Euler circuit.

Euler circuit: a simple circuit that contains every edge of the graph exactly once, starting and ending at the same vertex.

Condition: A connected graph has an Euler circuit $\Leftrightarrow$ every vertex has even degree.

Every vertex even-degree $\Leftrightarrow$ Euler circuit exists

Why: passing through a vertex uses edges in pairs (one to arrive, one to leave), so all degrees must be even.



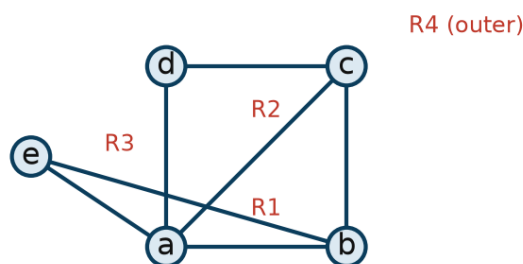
G_1 has an Euler circuit (all degrees even); G_2 has neither; G_3 has an Euler path only.

Q3. Prove that in a simple graph, the number of vertices of odd degree is always even.

Proof. Let V_1 = vertices of even degree, V_2 = vertices of odd degree. By the Handshaking theorem, $\sum_{v \in V} \deg(v) = 2e$, which is even. The sum over V_1 (a sum of even numbers) is even, so the sum over V_2 must be even too. A sum of odd numbers is even only when the count of terms is even. Hence the number of odd-degree vertices is even. {Number of odd-degree vertices is even} $\square$

Q4. State and prove Euler's formula for a connected planar simple graph.

Statement. For a connected planar graph with v vertices, e edges and r regions, $v - e + r = 2$.

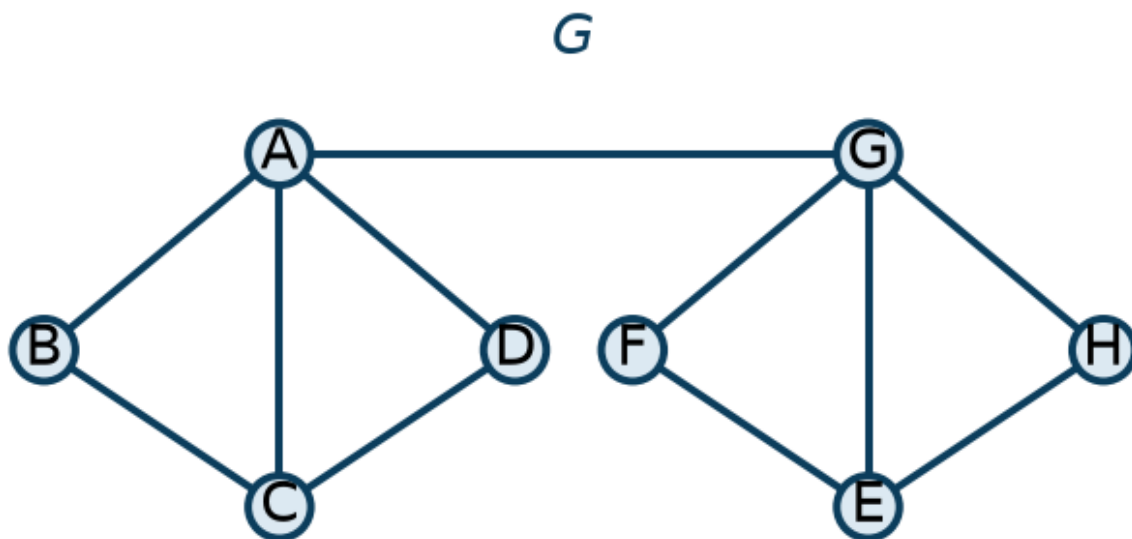


A planar graph splitting the plane into regions R_1, R_2, R_3, R_4 .

Proof (induction on edges). Base: one edge, $2-1+1=2$. Step: adding an edge either splits a region ($r\{+1, e\{+1$) or adds a new pendant vertex ($e\{+1, v\{+1$); in both cases $v-e+r$ is unchanged at 2. Hence $v-e+r=2$ always. $\{v-e+r=2\}$ $\square$

Q5. Use Fleury's algorithm to construct an Euler path or circuit in the graph shown.

Steps. (1) Check degrees: all even $\rightarrow$ Euler circuit (start anywhere); exactly two odd $\rightarrow$ Euler path (start at an odd vertex). (2) Move along a non-bridge edge unless no other edge remains. (3) Delete each edge after use. (4) Continue until all edges are used.



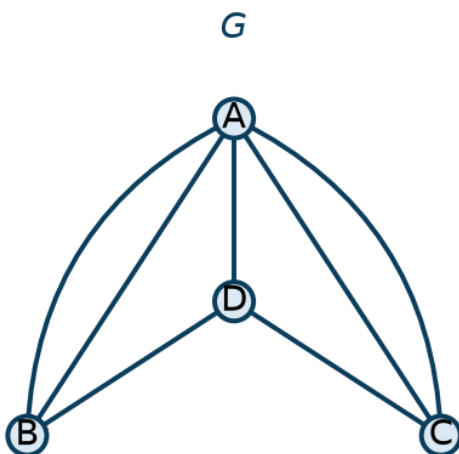
Graph for Q5 — degrees: $A=4, B=2, C=4, D=2, E=4, F=2, G=4, H=2$ (all even).

Apply: Every vertex has even degree, so an Euler circuit exists; start at A. Take A-B, B-C, C-D, D-A, A-C (keeping the bridge A-G for last), then C-E, E-F, F-G, G-E, E-H, H-G, and finally G-A.

Euler circuit: A-B-C-D-A-C-E-F-G-E-H-G-A

Q6. Check whether the graph shown has an Euler circuit or not.

Method. Compute the degree of every vertex. If the graph is connected and all degrees are even, it has an Euler circuit; if exactly two are odd, it has an Euler path but no circuit; if more than two are odd, it has neither.



Graph for Q6 — a multigraph with two edges each between A,B and between A,C, plus AD, BD, CD.

Apply: $\deg(A)=5, \deg(B)=3, \deg(C)=3, \deg(D)=3$. All four vertices have **odd** degree.

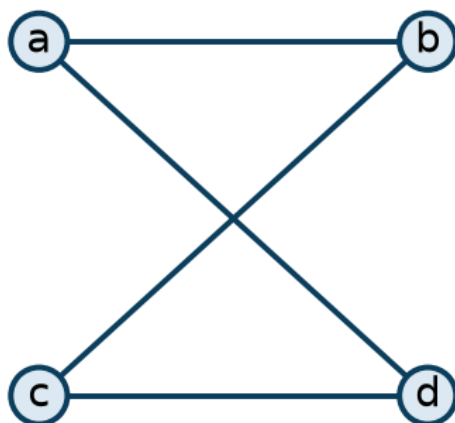
No Euler circuit (and no Euler path) — more than two vertices are of odd degree

Q7. Prove that every planar graph has a vertex of degree at most 5.

Same as Corollary 2 in Section 11.4. For a connected simple planar graph, $e \leq 3v-6 \Rightarrow 2e \leq 6v-12$. If all degrees were ≥ 6 , Handshaking would give $2e \geq 6v$, a contradiction. Hence some vertex has degree ≤ 5 . $\{v: \deg(v) \leq 5\} \neq \emptyset$ $\square$

Q8. Construct the adjacency matrix of the graph shown. How many paths of length four are there from a to d?

Adjacency matrix. Order the vertices a,b,c,d; entry (i,j) is 1 if the two vertices are joined by an edge, else 0. The graph is the 4-cycle a,b,c,d,a.



Graph for Q8 — a 4-cycle a,b,c,d,a.

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}.$$

Key idea. The number of paths of length k from v_i to v_j equals the (i,j) entry of A^k ; so we need

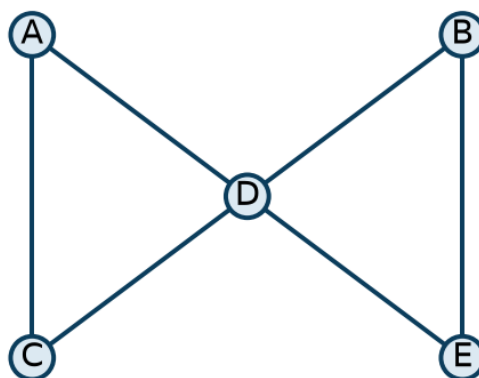
the (a,d) entry of A^4 . $A^4 = \begin{pmatrix} 8 & 0 & 8 & 0 \\ 0 & 8 & 0 & 8 \\ 8 & 0 & 8 & 0 \\ 0 & 8 & 0 & 8 \end{pmatrix}.$

The (a,d) entry is 0. Reason: in a 4-cycle a and d are in opposite colour classes, and a walk of **even** length 4 must end in the same class it started from — so it can never reach d.

0 paths of length four from a to d

Q9. Define an Euler circuit. Use Fleury's algorithm to find the Euler circuit in the graph shown.

Definition as in Q2. Apply Fleury: verify all degrees are even (so a circuit exists), start at any vertex, never take a bridge unless forced, and delete edges as used.



Graph for Q9 — two triangles sharing the vertex D; degrees: $A=2, B=2, C=2, E=2, D=4$ (all even).

Apply: Start at A: take A-C, C-D, D-B, B-E, E-D, and finish with D-A.

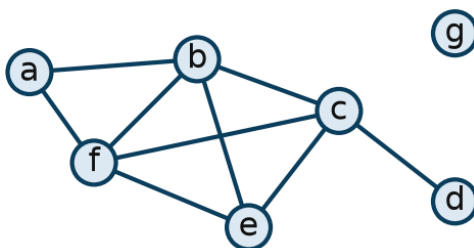
Euler circuit: A-C-D-B-E-D-A

Q10. Prove that a connected simple planar graph has a vertex of degree not exceeding 5.

Identical argument to Q7 / Corollary 2: $e \leq 3v-6 \Rightarrow 2e \leq 6v-12$; if all $\deg(v) \geq 6$ then $2e \geq 6v$, a contradiction. So at least one vertex has degree ≤ 5 . $\square$

Q11. Define the degree of a vertex in a graph.

The degree of a vertex is the number of edges incident with it; a **loop contributes 2** to the degree. It is denoted $\deg(v)$, and the sum of all degrees equals $2e$ (Handshaking theorem).

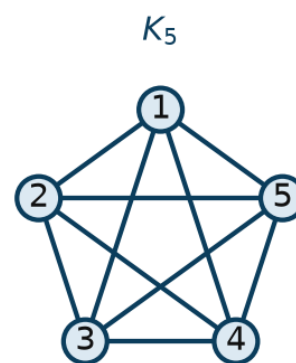
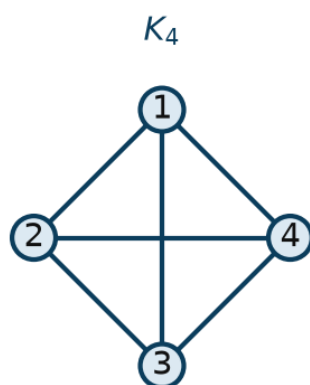
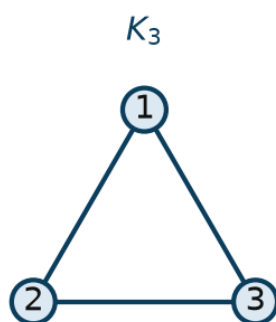


Here $\deg(a)=2, \deg(b)=4, \deg(c)=4, \deg(d)=1, \deg(e)=3, \deg(f)=4, \deg(g)=0$.

Q12. What is the chromatic number of a complete graph with 4 vertices?

In K_4 every pair of vertices is adjacent, so all four must receive different colours.

$\chi(K_4) = 4$

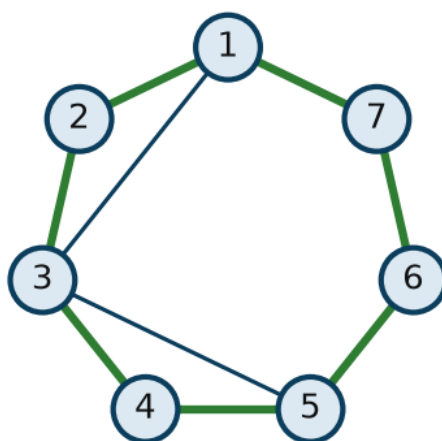


Complete graphs K_3, K_4, K_5 — $\chi(K_n)=n$.

Q13. What is a Hamiltonian circuit?

A cycle in a connected graph that contains **all** the vertices of the graph (each vertex exactly once, except the start, which is also the end). A graph containing such a circuit is called Hamiltonian.

Hamiltonian circuit (green)



A Hamiltonian circuit (shown in green) visits every vertex once and returns to the start.

Q14. A graph has 21 edges, 3 vertices of degree 4 and all other vertices of degree 2. Find the total number of vertices.

Set-up. Let the total number of vertices be n . Then 3 vertices have degree 4 and $(n-3)$ vertices have degree 2.

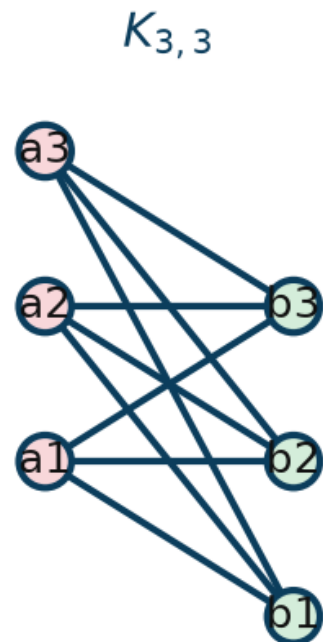
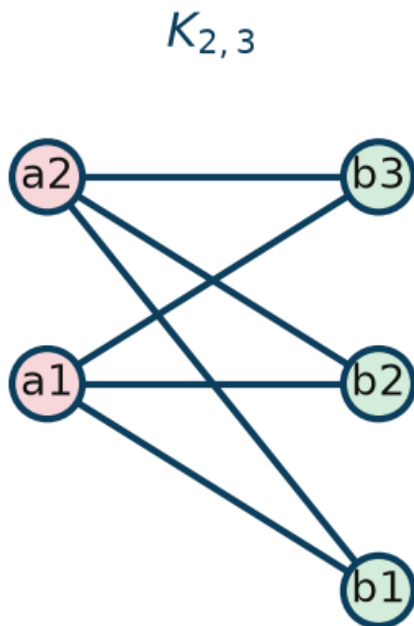
Handshaking: $\sum \deg = 2e = 2(21) = 42$. $3(4) + (n-3)(2) = 42 \Rightarrow 12 + 2n - 6 = 42 \Rightarrow 2n = 36 \Rightarrow n = 18$.

Total vertices = 18

Q15. How many edges does $K_{2,6}$ contain?

For a complete bipartite graph $K_{m,n}$ the number of edges is $m \times n$. Here $2 \times 6 = 12$.

12 edges

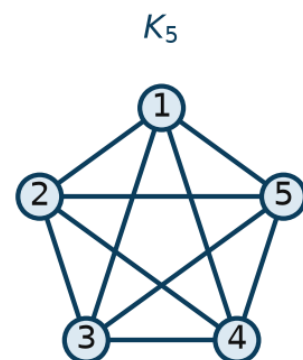
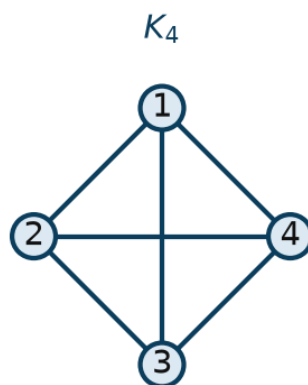
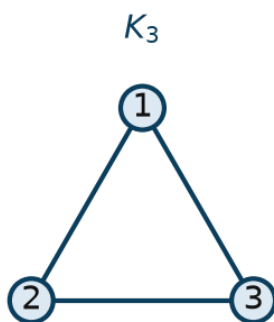


Complete bipartite graphs $K_{2,3}$ and $K_{3,3}$ — $|E(K_{m,n})| = mn$.

Q16. Find the chromatic number of the complete graph K_5 .

All 5 vertices are mutually adjacent, so all need distinct colours.

$\chi(K_5) = 5$



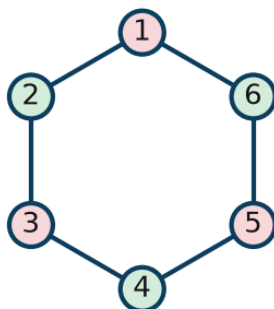
K_5 needs 5 colours.

Q17. What is the chromatic number of an odd cycle?

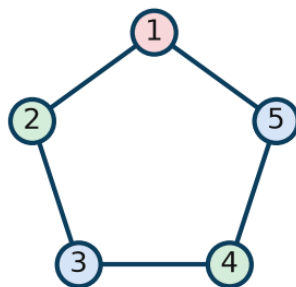
An odd cycle cannot be 2-coloured (that would require it to be bipartite, i.e. to have no odd cycle), so it needs exactly 3 colours.

$\chi(C_n) = 3$ for odd n

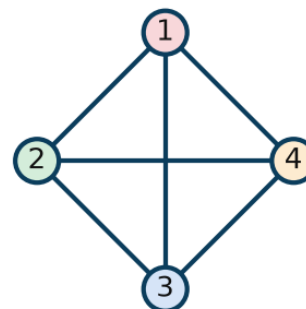
$C_6: \chi = 2$



$C_5: \chi = 3$



$K_4: \chi = 4$



C_6 needs 2 colours, C_5 (odd) needs 3, and K_4 needs 4.

Q18. How many edges does K_{100} contain?

$|E(K_n)| = C(n, 2) = n(n-1)/2$. So $|E(K_{100})| = (100 \times 99)/2 = 4950$.

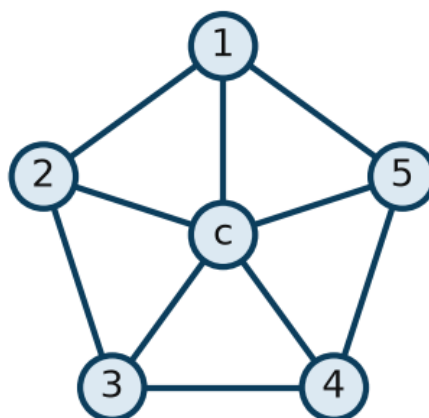
4950 edges

Q19. How many edges does W_{100} contain?

W_n has n rim edges plus n spokes, i.e. $2n$ edges. So $|E(W_{100})| = 2 \times 100 = 200$.

200 edges

Wheel W_5



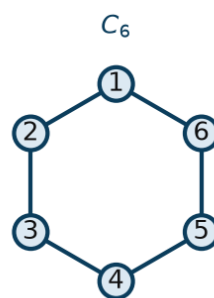
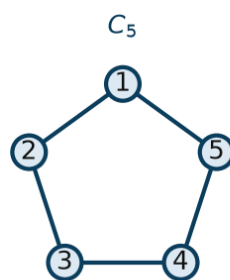
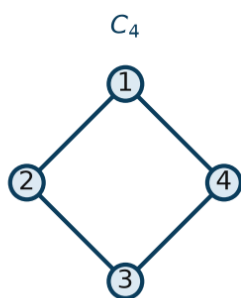
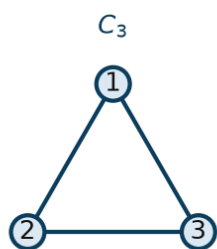
The wheel W_5 — a cycle plus a hub joined to every rim vertex.

Q20. Determine which of the following graphs is bipartite: (a) C_3 (b) C_5 (c) C_4 (d) C_6 .

A cycle is bipartite iff its length is even (no odd cycle).

- (a) C_3 — odd → **not bipartite**
- (b) C_5 — odd → **not bipartite**

- (c) C_4 — even $\rightarrow$ **bipartite** ✓
- (d) C_6 — even $\rightarrow$ **bipartite** ✓



Cycles C_3, C_4, C_5, C_6 .

Bipartite: C_4 and C_6

14. Quick Revision Summary

Definitions to Remember

- **Graph** $G=(V,E)$; **loop** (edge from a vertex to itself); **parallel edges**.
- **Simple** (no loops/parallel), **multigraph** (parallel allowed), **pseudograph** (loops + parallel), **digraph** (directed edges).
- **Degree** = number of incident edges; a loop counts 2. **Isolated** vertex: degree 0. **Pendant** vertex: degree 1.
- **Adjacent** vertices / **incident** edge; **neighbourhood** $N(v)$.
- **Path**, **circuit**, **simple path**; **connected** graph; **cut vertex**; **bridge**.
- **Euler circuit/path** (covers every edge once); **Hamiltonian circuit/path** (covers every vertex once).
- **Planar graph**, **region/face**; **bipartite**; **chromatic number** $\chi(G)$.

Formulas

Concept	Formula
Handshaking theorem	$2e = \sum \{v\} \deg(v)$
Edges in K_n	$n(n-1)/2$
Edges in $K_{m,n}$	mn
Edges in W_n	$2n$
Euler's formula (connected planar)	$v-e+r=2$
Euler's formula (k components)	$v-e+r=k+1$
Simple planar bound	$e \leq 3v-6$
Triangle-free planar bound	$e \leq 2v-4$
Paths of length k from v_i to v_j	(i,j) entry of A^k

Key Theorems & Results

- Number of odd-degree vertices is **even**.
- **Euler circuit** exists iff connected and all degrees even; **Euler path** exists iff exactly two odd-degree vertices.
- Every connected simple planar graph has a vertex of degree ≤ 5 .
- K_5 and $K_{3,3}$ are **not planar**.
- A graph is **bipartite iff no odd cycle**; equivalently, 2-colourable.
- **Four-colour theorem**: every planar graph has $\chi \leq 4$.

Chromatic Numbers

Graph	χ
-------	--------

No edges	1
Path / tree / even cycle / bipartite	2
Odd cycle C_n	3
Even wheel W_n	3
Odd wheel W_n	4
Complete graph K_n	n
Any planar graph	≤ 4

— End of Unit III (Graph Theory) Notes —